

Project 2: Coffee Shop Retail Sales Analysis

1. Project Objective The goal of this project was to analyze transactional and footfall data across three coffee shop locations (Astoria, Hell's Kitchen, and Lower Manhattan). The objective was to build an interactive dashboard tracking key performance indicators (KPIs) like peak hours, top-selling products, and regional performance to optimize staffing and inventory.

2. Technical Skills Demonstrated

- **Dashboard Design & Theming:** Created a visually cohesive, branded dashboard using custom color palettes to enhance user experience and readability.
- **KPI Development:** Built summary cards to highlight critical top-line metrics (Total Sales, Total Footfall, Average Sales Per Customer, and Average Orders Per Customer).
- **Time-Series Analysis:** Utilized line charts and column charts to map out customer purchasing behavior by hour of the day and day of the week.
- **Interactive Controls:** Implemented slicers (Size, Day Name, Month Name) to enable deep-dive filtering for specific operational insights.

3. Key Business Insights

- **Peak Operational Hours:** Customer traffic heavily spikes in the morning, peaking sharply between 8:00 AM and 10:00 AM, before leveling out for the rest of the workday.
- **Location Parity:** All three store locations perform remarkably consistently. Hell's Kitchen leads slightly with \$236K in sales, but Astoria (\$232K) and Lower Manhattan (\$230K) see nearly identical footfall and revenue.
- **Product Preferences:** Coffee dominates the sales mix (39%), closely followed by Tea (28%). The "Barista Espresso" is the standout top-grossing product, generating over \$91,000 in revenue alone.
- **Weekday vs. Weekend Trends:** Order volume remains remarkably steady throughout the work week (Monday–Friday), with a noticeable dip in footfall on Saturdays and Sundays.

4. Strategic Recommendations

- **Optimized Staffing:** Allocate the most experienced staff and maximize employee headcount during the critical 7:00 AM – 10:00 AM morning rush to reduce wait times and capture maximum revenue.
- **Cross-Selling Opportunities:** Since Bakery items only account for 12% of sales, introduce automated upsell prompts (e.g., pastry pairings) at the register specifically for customers ordering the top-selling Barista Espresso or Brewed Chai.
- **Weekend Promotions:** Introduce weekend-specific loyalty perks or specialty drinks to boost the lower footfall observed on Saturdays and Sundays.